# Sales Regression Model

## Predicting Sales Using Machine Learning

### 1. Objective

The objective is to build a regression model that predicts Sales from historical transaction, customer, product, channel, pricing, discount, marketing and time-based information.

### 2. Data Preparation

The dataset contains 6,645 records. The Date field was converted to datetime and additional Year, Month, Quarter and DayOfWeek features were created. Missing MarketingSpend values were replaced using median imputation. Categorical variables were one-hot encoded and numerical variables were median-imputed inside the modeling pipeline.

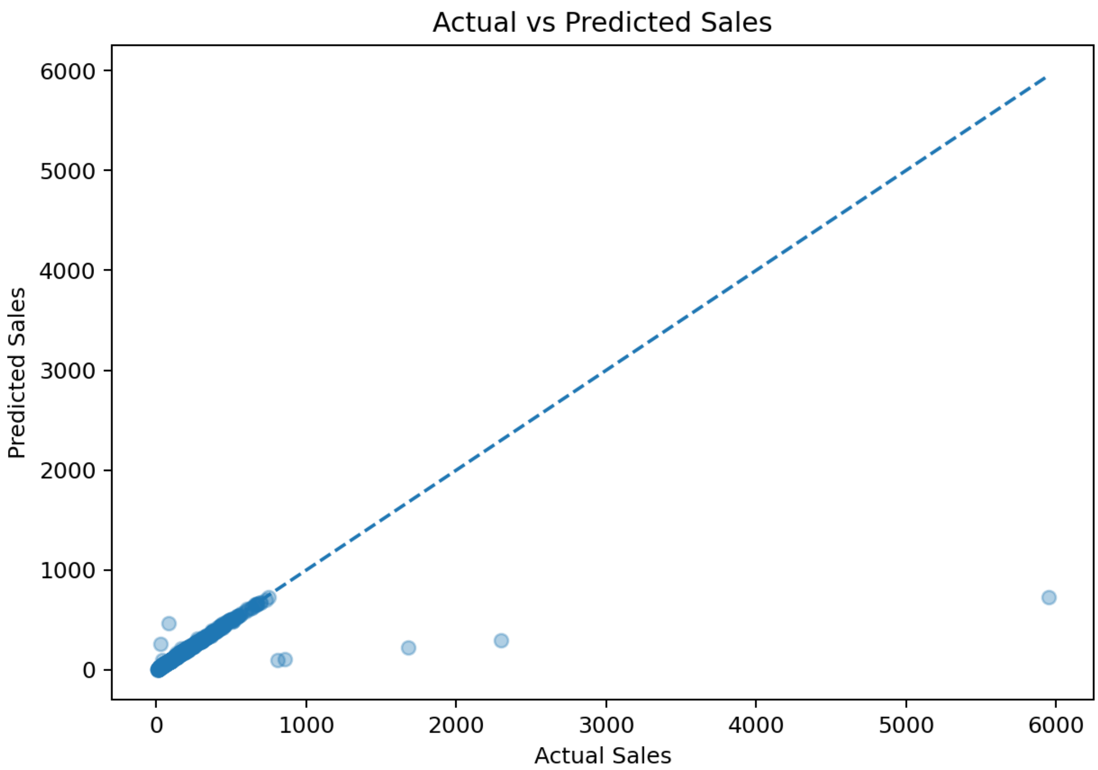
### 3. Model Used

**Gradient Boosting Regressor** was selected because it can capture nonlinear relationships and interactions between sales drivers while handling a mixture of numerical and categorical features after preprocessing.

| Parameter | Value |
|---|---|
| Algorithm | Gradient Boosting Regressor |
| Estimators | 200 |
| Learning Rate | 0.05 |
| Maximum Depth | 3 |
| Random State | 42 |
| Train/Test Split | 80% / 20% |

### 4. Model Performance

| Metric | Result | Interpretation |
|---|---|---|
| $R^2$ | 0.442 | Explains about 44% of test-set sales variance |
| MAE | 12.37 | Average absolute prediction error |
| RMSE | 161.71 | Penalizes larger prediction errors more strongly |
| Training Records | 5,316 | 80% of available records |
| Testing Records | 1,329 | 20% held out for evaluation |




## 5. Interpretation

The model provides a useful baseline for sales prediction, with an $R^2$ of approximately 0.44. The difference between MAE and RMSE indicates that some observations have substantially larger prediction errors. These may represent high-value or unusual transactions and should be investigated before using the model for operational forecasting.

## 6. Recommended Improvements

• Perform systematic outlier analysis and investigate unusually large sales transactions.

• Add lagged sales, rolling averages and seasonal features if historical time-series forecasting is required.

• Create campaign-level marketing features to measure marketing ROI more accurately.

• Compare Gradient Boosting with Random Forest, XGBoost or other boosting approaches.

• Use cross-validation and hyperparameter tuning before production deployment.

• Evaluate the model separately across regions, categories and channels to identify performance gaps.

## 7. Conclusion

A complete regression pipeline was developed to predict Sales from the available business data. The pipeline includes preprocessing, feature engineering, model training and evaluation, making it suitable as a professional baseline for the sales analytics capstone project.